

Project Overview

1. The Problem

Doctors write clinical notes, but hospitals need to assign medical codes (ICD codes) to these notes for billing and research. Right now, people manually read each note and decide which codes apply—this is slow and makes mistakes.

2. Your Task

You need to build a machine learning model that automatically identifies whether a sentence in a clinical note contains useful medical information that could be assigned an ICD code. In other words: Does this sentence contain ICD-codable medical information? Yes or No.

3. What You Get

- A `train\_data-text\_and\_labels.csv` file containing 20 example sentences from clinical notes and their metadata from MIMIC-III database that are already labeled as "useful" or "not useful"
- Several `testX\_text\_only.csv` files containing sentences from clinical notes where you will need to label (have your model make predictions on).
- The entire MIMIC-III dataset (as you've become a credentialed user after completing HW1).

4. What You Need to Build

A machine learning model that can:

- Work offline (no internet or external APIs allowed)
- Accept any text input up to 128 words and predict whether it contains ICD-codable medical information

5. What's Up to You

The following design decisions are up to you, and you are required to document them:

- How you prepare and segment the data
- What machine learning approach do you use (logistic regression, neural networks, pre-trained language models, etc.)
- How you train and test your system
- You can also use additional unlabeled examples from MIMIC-III or other public datasets to train your model

DO NOT UPLOAD THIS OR MIMIC-III DATA TO OpenAI/ChatGPT/Generative AI resources that ingest data. Doing so is a direct violation of course policies!

Project Deliverables (100 points)

The final project contains several portions, and you will complete the following materials for the assignment:

- You will build an ML model that predicts whether a medical sentence is associated with an ICD code.
- You will submit your predictions to Gradescope and pass the test cases.
- You will write a conference-paper–style report that documents everything you did.
- You will prepare a presentation about your work.

The final project submission is divided into three submission portals:

- A “Final Project Code” submission portal that links to the Gradescope autograder.
- A “Final Project Report” submission portal where you will submit a PDF file.
- A “Final Project Code, Source, Presentation, and Video Turn-In” submission portal where you will submit a zipped folder.

1. Programming Portion (60 points)

Gradescope Autograder submissions are **individual**—each team member (if applicable) must submit separately.

Complete this portion by:

- Submitting your model’s prediction (test##-pred.csv) CSV files to Gradescope and passing the test cases. Refer to the Gradescope autograder for the point distribution and test cases.
- Adding your source code to a zipped file and submitting it to Canvas after you complete all portions.

2. Written Portion (25 points)

Write a scientific report in at least 4 pages in the IEEE double-column format using this template (<https://www.ieee.org/conferences/publishing/templates.html>). The paper must include:

- **Introduction**
 - Briefly describe the problem space and why it matters.
 - Introduce prior works in the field and include relevant citations.
- **Methods**
 - Explain your approach and any references that informed it.
 - Explain the technical details of your implementation.
 - Include figures that illustrate your method.
- **Experiments & Results**
 - Describe what you tested and explain why.
 - Present results using figures or tables.
- **Discussion**
 - Discuss the significance of your results
 - Discuss any blockers and obstacles you’ve encountered
- **Generative AI Disclosure**
 - Disclose whether and where did you used generative AI to complete the project.
 - If you used generative AI to help you code, then report:
 - The prompts you used.
 - Mistakes the AI made that you identified.
 - How you validated correctness (beyond the autograder).
- **Conclusion**
 - Summarize key findings and takeaways.
 - Disclose each member’s contribution to your team (if applicable).

Complete this portion by:

- Submitting your written report as a PDF to Canvas Turnitin.
- Adding your LaTeX source code to the same zipped file containing your source code.

3. Presentation Portion (15 points)

Prepare and record a **15-minute presentation** summarizing your final project. Your presentation should include:

- **Motivation & Problem Statement**
 - Clearly explain the problem you addressed and why it matters.
 - Provide any necessary background for understanding your approach.
- **Methods Overview**
 - Summarize your approach and highlight the core technical ideas.
 - Include diagrams or visual aids that help illustrate your method.
- **Experiments & Key Results**
 - Present the experiments you conducted and the rationale behind them.
 - Show your most important results using clear figures or tables.
 - Briefly interpret what the results mean.
- **Discussion**
 - Highlight key insights, strengths, and limitations of your approach.
 - Mention any surprising findings or challenges encountered.
- **Conclusion**
 - Summarize the main takeaways of your project.
 - Optionally, mention potential extensions or future work.
- **Delivery Requirements**
 - Presentation must be within **15 minutes** and each team member (if working as a team) must present at least one slide.
 - Exported your presentation as PPTX file for submission.
 - Slides should be visually clear, well-organized, and professional.

Complete this portion by:

- Recording your presentation as a video (do it inside PowerPoint), then adding the presentation PPTX file containing the embedded video to the same zipped file.

Good luck and start early!